

Palindrome Checker (Integer)

Java · digit reversal · number theory

Description

This program checks whether a given integer is a palindrome — that is, whether it reads the same forwards and backwards. It works by reversing the digits of the number mathematically and comparing the reversed value to the original. Negative numbers are automatically treated as non-palindromes.

Source Code

```
import java.util.Scanner;

public class PalindromeChecker {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Take input from user
        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        int original = number;
        int reverse = 0;

        // Negative numbers are not considered palindrome
        if (number < 0) {
            System.out.println("Palindrome: false");
        }
        else {

            // Reverse the integer
            while (number != 0) {
                int digit = number % 10;
                reverse = reverse * 10 + digit;
                number = number / 10;
            }

            // Compare original number with reversed number
            if (original == reverse) {
                System.out.println("Palindrome: true");
            }
            else {
                System.out.println("Palindrome: false");
            }
        }

        sc.close();
    }
}
```

```
}  
}
```

How It Works

1. The user enters an integer, which is stored in 'number' and copied into 'original'.
2. If the number is negative, it is immediately reported as not a palindrome.
3. Otherwise, the digits are extracted one at a time using the modulus operator (% 10) and used to build 'reverse'.
4. Once all digits are processed, 'original' is compared to 'reverse'.
5. The result is printed as true or false.

Time Complexity: $O(d)$ where d = number of digits | Space Complexity: $O(1)$